

Ruben IFRAH

M.Sc. Student in AI – École Polytechnique & Dauphine-PSL

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Education

Dauphine - PSL <i>M2 IASD (AI, Systems, Data)</i>	2025 – Present <i>Paris, France</i>
• Ranked 3rd in the cohort • Highly selective research-oriented Master's program co-accredited by ENS-Ulm • Management committee including Olivier Cappé & Christian P. Robert	
École Polytechnique <i>Ingénieur Polytechnicien diploma</i>	2021 – 2025 <i>Palaiseau, France</i>
• Majored in Applied Maths and Statistics • Economics Minor: econometrics, financial models, micro/macro economy • Strong scientific background: general physics, quantum physics, fluid mechanics, solid mechanics, econo-physics • GPA: 3.7/4.0	
Lycée Saint-Louis <i>CPGE PCSI-PSI*</i>	2019 – 2021 <i>Paris, France</i>
• A two-year intensive program in physics, mathematics and engineering sciences • Ranked 1st in the 2020-2021 PSI* cohort	

Experience

Clevaligner <i>3D Vision Scientist Intern</i>	June 2025 – Aug. 2025 <i>Paris, France</i>
• Worked in a startup on AI applied to the medical field (dental correction). • Responsible for assessing and implementing SOTA models in point-cloud teeth segmentation (Transformer-based PointNet). • Assisted AI engineers in designing and constructing point-cloud pipelines.	
University of Valencia <i>Research Assistant (Statistics)</i>	April 2024 – July 2024 <i>Valencia, Spain</i>
• Worked on ecological inference estimators and statistical modeling. • Developed an extended model using different norms for estimator efficiency.	
MBDA <i>Applied Research Intern</i>	May 2023 – Aug. 2023 <i>Plessis-Robinson, France</i>
• Signal Processing & Optimization • Designed anti-jamming filters for radar systems dealing with stochastic noise. • Modeled physical constraints using MATLAB simulations to validate robustness.	
French Army (Saint-Cyr) <i>Officer Cadet / Platoon Leader</i>	Sept. 2021 – April 2022 <i>Martinique & Coëtquidan</i>
• Platoon Leader (Jan 2022 - April 2022): Led a platoon of 30 new recruits soldiers in Martinique. Organized training exercises and managed logistics. • Military Academy (Sept 2021 - Jan 2022): Intensive officer training at Saint-Cyr Coëtquidan. Developed leadership, rigor, and team management skills.	

Objective

I am an **Ingénieur Polytechnicien** (X21) currently pursuing the **M2 IASD** (AI, Systems, Data) at Dauphine-PSL. With a background in Physics and **Applied Mathematics**, I am seeking opportunities in **Deep Learning**, with a strong interest in Deep-RL and Computer Vision.

Projects

[Portfolio](#)

Deep RL Autonomous Racing Agent	
• Designed a custom LIDAR-based environment with Procedural Generation (Harmonic Noise) to enforce policy generalization and prevent overfitting. Implemented Dueling DDQN from scratch, demonstrating superior stability by decoupling State and Values streams.	
Geometric Shortcuts in 3D SSL	(Ongoing Research)
• Studying the emergence of geometric shortcut representations in large-scale 3D self-supervised learning, evaluating whether pretext objectives induce feature collapse toward low-level spatial cues.	
Bayesian Physics-Informed NN	
• Evaluated uncertainty calibration in Bayesian PINNs using Hamiltonian Monte Carlo. Extended the framework with dynamic noise inference and analyzed the spectral bias limitations of its infinite-width NNGP limit	
DPO alignment for Mistral-7B	
• Implemented and analyzed Direct Preference Optimization on the Anthropic-HH dataset. Achieved 58% win-rate vs SFT baseline. Analyzed the impact of beta parameter on reward collapse	

Skills

Deep Learning & AI:
• PyTorch, Gymnasium, JAX, Hugging Face, WandB
Dev & Scientific:
• Git, Docker, Linux (Bash), LaTeX, MATLAB, R, SQL
Languages:
• French (Native), English (C1 - Fluent), Spanish (B1)

Selected Coursework (M2 IASD)

Learning Theory & Optimization
• Foundations of Machine Learning (<i>F. Bach</i>) 16/20
• Math of Deep Learning (<i>B. Loureiro</i>)
• Non-Convex Inverse Problems (<i>I. Waldspurger</i>)
• Optimization for ML (<i>C. Royer</i>) 15.25/20
Reasoning, RL & GenAI
• Reinforcement Learning (<i>O. Cappé</i>) 15/20
• Monte Carlo Search (<i>T. Cazenave</i>)
• Large Language Models (<i>A. Allauzen</i>) 17.6/20
• LLM for Code & Proofs (<i>M. Lelarge</i>)
Geometric DL & Computer Vision
• Manifold Learning (<i>E. Aamari</i>)
• Point Clouds & 3D Modeling (<i>J.E. Deschaud</i>)
• Deep Learning for Image Analysis (<i>T. Walter</i>) ... 17.8/20